BLOOD DONATION SYSTEM

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Project Overview

Blood donation is a critical service, yet many patients struggle to find donors in emergencies due to a lack of organized communication. This system is a centralized platform designed to bridge the gap between voluntary blood donors and those in urgent need.

By automating the donor search and notification process, this system aims to save lives through faster response times.

Project Motivation

- Patients often struggle to find donors in emergencies due to a lack of organized communication and reliance on manual record-keeping.
- Replacing traditional methods with a real-time database that makes help available in just a few clicks.
- Digitizing the process to address a vital social need and minimize the time spent searching for donors.

Features

User & Donor Management

- Registration and login for donors and recipients to manage their personal profiles.

Smart Search

- A search bar to filter available donors based on blood group and specific city locations.

Emergency Postings

- A public board on the homepage where recipients can post and view urgent blood requirements.

Admin Dashboard

- Tools for administrators to verify donor information and manage system data to ensure validity.

Technology

Frontend	Backend	Database	Version Control & Deployment
React.js for building a dynamic, mobile-responsive, and user-friendly interface.	**Node.js** and **Express** for fast and efficient server-side processing.	**MongoDB** for flexible, real-time data storage.	Managed via **Git/GitHub** and deployed on platforms like **Vercel**

Future Scope

- **Automated Alerts**: Implementing Email or SMS notifications to alert nearby donors automatically when an emergency request is posted.

- **Eligibility Tracker**: A system to calculate the next available donation date based on a donor's last submission history.

Medical Integration: · Special thanks to Shondhani Blood Bank Osmani Medical College, for providing valuable data and support.

Conclusion

"Blood Donation System" digitizes a life-saving service to make blood donation more accessible and organized. The project provides hands-on experience in full-stack development and database management. A reliable platform committed to saving lives through technology.

THANK YOU!

DONATE BLOOD